

Anum Fatima

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RESEARCH EXPERIENCE

PDRA, Department of Statistics, University of Oxford (2026-present)

- Auditing differentially private graph generators
- Develop differentially private graph generators based on SteinGen

DPhil Research, Department of Statistics, University of Oxford (2021-2026)

- Developed a Stein characterisation for inhomogeneous random graph (IRG) models, enabling rigorous comparison of network models via common underlying processes
- Developed a kernelised Stein discrepancy test, IRG-gKSS, for assessing the fit of IRG models, with theoretical guarantees, and applications in detecting lack of fit of IRG models to network data
- Extended Stein's method to competing and complementary risks (CCR) models, deriving bounds between CCR distributions and approximating families, thereby enhancing the interpretability of complex systems through simpler probabilistic frameworks

Research Assistant, FAIR project at University of Oxford (12-week contract, 2025)

- Applied the IRG-gKSS framework to detect structural anomalies in network data, further validating its performance on real-world and synthetic networks.
- Designed experiments to evaluate the power of IRG-gKSS under varying network conditions, using different graph kernels.

Independent research (2013-2015)

- Developed new probability distributions, including the extended Poisson exponential distribution and the transmuted exponentiated Pareto-I distribution
- Analysed and established their key distributional properties and potential applications

Master's Dissertation, Department of Statistics, Lahore College for Women University (2011-2013)

- Developed the generalised Poisson exponential distribution and conducted a detailed study of its distributional properties

RELEVANT RESEARCH SKILLS

- Expertise in Stein's method for characterisation of probability distributions
- Development of kernelised Stein discrepancy type tests applied to network models and graph data
- Comparing and using different graph kernels to measure network similarities
- Extensive experience in simulation and Monte Carlo testing frameworks using R programming language
- Ability to translate complex mathematical tests and methodologies into efficient and practical R code implementations
- Skilled in formulating, developing, and analysing compounded and novel probability distributions

AWARDS & SCHOLARSHIPS

- Commonwealth PhD Scholarship (for least developed countries and fragile nations), University of Oxford (2020) – Covered full tuition fees, a living stipend, and a family support allowance
- International travel awards and grants to present research at the following events:
 - RandNET meeting – Workshop on Graph Limits and Networks, Charles University, Prague (2023)
 - 43rd Conference on Stochastic Processes and their Applications, University of Lisbon (2023)
 - Distance-Based Methods in Machine Learning Workshop, University College London (2023)
 - UK Easter Probability Meeting, University of Manchester (2023)
- Honorarium for online presentation at the Stein's Method: The Golden Anniversary, National University of Singapore (2022)

- University Gold Medals, roll of honours, and cash prizes awarded for exceptional academic performance in MS (2013) and BS (2011) degrees
- Merit-based undergraduate scholarship, awarded annually (2007–2011)

EDUCATION

University of Oxford, Jan 2021 – Present
 DPhil in Statistics: Stein characterisation of distributions with a particular emphasis on Bernoulli mixture graph models; dissertation submission planned for September 2025.

Lahore College for Women University, Jan 2011 – 2013
 MS in Statistics: Generalised Poisson exponential distribution.

Lahore College for Women University, Jan 2007 – 2011
 BS in Statistics: Distributional properties of generalised order statistics for extended exponential distributions.

TEACHING EXPERIENCE

Tutor, University of Oxford (2023-2024)

- Delivered tutorial sessions for the *Probability and Statistics for Network Analysis* course, to groups of up to 13 students
- Assessed student progress and submitted end-of-term reports via the Oxford Teaching Management System (TMS)

Graduate Teaching Assistant, University of Oxford (2021-2023)

- Marked problem sheets and provided constructive feedback for the *Probability and Statistics for Network Analysis* and *Applied Probability* courses
- Managed attendance records and problem sheet grading using TMS.
- Led a demonstration of one problem from the problem sheet in each tutorial session

Lecturer in Statistics, Lahore College for Women University, Lahore (2017-2020)

- Taught undergraduate statistics courses to groups of up to 50 students
- Took responsibility for reviewing and updating course outlines within university guidelines
- Participated in curriculum development and academic planning with departmental committees
- Prepared and marked final exams, coursework, and in-term assessments
- Supervised undergraduate dissertations and internship reports
- Served on departmental academic, admissions, and examination committees

Mentor/Supervisor, Lahore College for Women University and Queen Mary College, Lahore (2017-2020)

- Supervised undergraduate theses on the following topics:
 - Factors affecting the choice of buyers to shop from shopping malls or local markets
 - Satisfaction of youth on new government policies: a cross-sectional study at LCWU, GCU, PU and KC Lahore
- Supervised multiple internship reports for students in the Department of Statistics

Lecturer in Statistics, Queen Mary College, Lahore (2015-2017)

- Delivered undergraduate statistics courses following centrally prescribed curricula
- Taught students from multiple departments in line with the district board's academic guidelines
- Conducted assessments, marked internal examinations, and provided regular feedback

Visiting Teaching Assistant, Lahore College for Women University, Lahore (2014-2015)

- Taught undergraduate statistics courses and contributed to exam preparation and marking
- Responsible for preparing and marking final exam papers
- Delivered statistics courses to year 11 and 12 students, managing classes of up to 50 students

ADMINISTRATIVE & LEADERSHIP ROLES

Member, Departmental Academic Committee

Working as a part of a team of 4 members,

- Maintained up-to-date course outlines and scheme of studies
- Supported curriculum development within university guidelines
- Organised departmental Board of Studies (BOS) meetings, liaising with internal and external members and preparing internal evaluation reports

Member, MS Admission Committee

Working as a team of two people,

- Contributed to setting admission criteria and preparing advertisement materials
- Scrutinised application forms and managed logistics for admission tests and interviews
- Prepared final merit lists and managed activities at the admissions desk

Member, Departmental Examination Committee

Working as part of a team of 6 members,

- Coordinated the collection and distribution of exam papers and award lists between the Statistics Department and the Examination Office

Apart from these responsibilities, I was also

Department Stationery In-Charge

- Managed inventory and procurement of stationery for the department, ensuring timely supply to faculty and staff

Trained Facilitator, Active Citizens Programme (British Council)

- Delivered bi-weekly training sessions and facilitated student-led community projects

PUBLICATIONS

Preprints

- Fatima, A., Reinert, G. (2025). *A Kernelised Stein Discrepancy for Assessing the Fit of Inhomogeneous Random Graph Models*. arXiv:2505.21580.
- Fatima, A., Reinert, G. (2025). *Stein's Method for Distributions Modelling Competing and Complementary Risk Problems*. arXiv:2212.09615. (Accepted for publication in *Advances in Applied Probability*)

Journal Papers

- Fatima, A., Roohi, A. (2016). *Transmuted Exponentiated Pareto-I Distribution*. *Pakistan Journal of Statistics*, 32(1), 63–80.
- Fatima, A., Roohi, A. (2015). *Extended Poisson Exponential Distribution*. *Pakistan Journal of Statistics and Operation Research*, XI(3), 361–375.
- Fatima, A., Roohi, A. (2015). *The Generalised Poisson Exponential Distribution*. *Journal of ISSOS*, 1(2), 103–118.
- Aziz, A., Roohi, A., Fatima, A. (2013). *Distributional Properties of Generalised Order Statistics for Exponentiated Pareto-I Distribution*. *International Journal of Applied Research*, 2, 62–75.

Conference Papers

- Fatima, A., Reinert, G. (2023). *Assessing the Fit of Erdős–Rényi Mixture Graph Models*. Extended abstract available in: *Complex Networks 2023: The 12th International Conference on Complex Networks & Their Applications*, Book of Abstracts.
- Ramzan, M., Fatima, A. (2019). *Determinants of Satisfaction of Youth on New Government Policies: A Cross-Sectional Study at LCWU, GCU, PU and KC Lahore*. *15th Islamic Countries Conference on Statistical Sciences*, Lahore, Pakistan (December 21–24), 34: 453–466. Pakistan: Islamic Countries Society of Statistical Sciences, Lahore.

- Fatima, A., Roohi, A. (2012). *Distributional Properties of Generalised Order Statistics for Extended Exponential Distribution*. 9th International Conference on Statistical Sciences, 22: 147–158. Lahore, Pakistan: Islamic Countries Society of Statistical Sciences.

RESEARCH DISSEMINATION

Oral Presentations

- *Assessing the Fit of Erdős–Rényi Mixture Models Using Stein’s Method*, Bernoulli–IMS 11th World Congress, Bochum, Germany (August 2024).
- *Assessing the Fit of Erdős–Rényi Mixture Models*, 12th International Conference on Complex Networks & Their Applications, Menton, France (September 2023).
- *Stein’s Method for Erdős–Rényi Mixture Graph Models*, RandNET Workshop, Charles University, Prague (September 2023).
- *Stein’s Method for Erdős–Rényi Mixture Graph Models*, Distance-Based Methods in Machine Learning Workshop, University College London, UK (June 2023).
- *Stein’s Method for Poisson–Exponential Distributions*, Stein’s Method: The Golden Anniversary, National University of Singapore (July 2022).
- Additional oral presentations at national and international conferences between 2012 and 2015.

Poster Presentations

- Presented on Stein’s method and Erdős–Rényi mixture graph models at multiple prestigious events including:
 - University of Oxford (2024)
 - University College London (2023)
 - University of Lisbon (2023)
 - UK Easter Probability Meeting (2023)
- Awarded first prize at poster competitions in Pakistan (2012, 2014).

ADDITIONAL SKILLS & TRAINING

Software Carpentry Workshop (Shell, Git, Python), University of Oxford (2025)

CUDA Programming on NVIDIA GPUs, University of Oxford (2024)

Equity, Diversity & Inclusion Training: Supporting Neurodivergent Staff, University of Oxford

PROFESSIONAL MEMBERSHIPS

Royal Statistical Society, Student Member (2021 – Present)